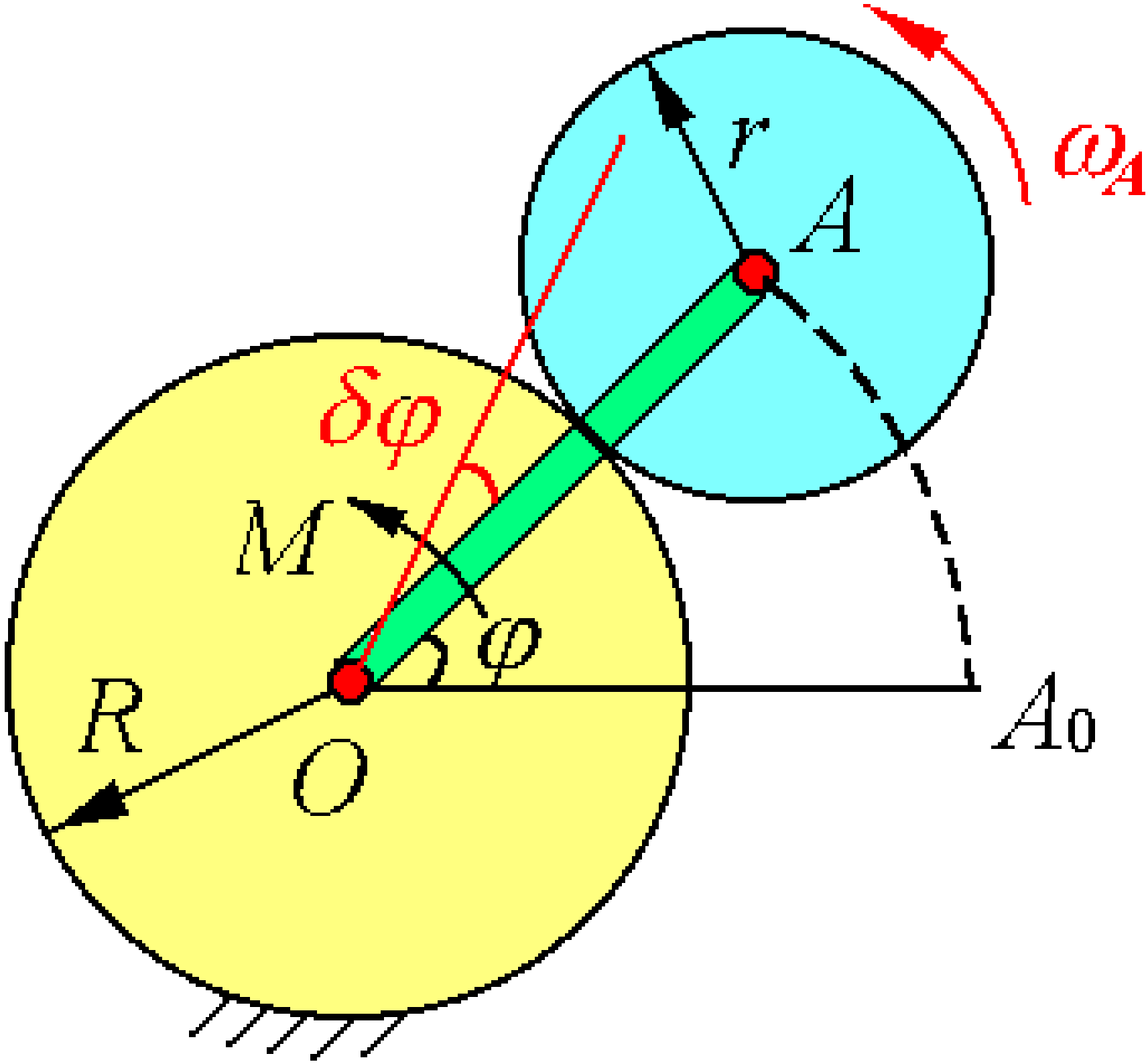

ω_A
r
A
$\delta\varphi$
M
φ
R
O
A_0